VQE Runtime ignores any parameters of optimizer other than SPSA - QNSI

https://github.com/Qiskit/qiskit-ibm-runtime/issues/407

ROW 112

VQE Runtime ignores any parameters of optimizer other than SPSA/QNSPSA #407

New issue

Closed | Qiskit/qiskit#8381

LenaPer opened on Jul 4, 2022

Describe the bug
When running a vqe runtime through QiskitRuntimeService(channel='ibm_quantum') with an optimizer other than SPSA/QNSPSA, and looking at the logs (job.logs()), I noticed the optimizer is set to be SLSQP, even when I pass any other optimizer, it shows on the logs :
`2022-07-04T09:37:35.046460730Z -- optimizer: <qiskit.algorithms.optimizers.slsqp.SLSQP object at 0x7fed33b72d60>`
And later in the logs :

```
2022-07-04T09:37:35.046639714Z ==============================================================
2022-07-04T09:37:35.046645335Z Optimizer: SLSQP
2022-07-04T09:37:35.046650890Z -- method: slsqp
2022-07-04T09:37:35.046656375Z -- bounds_support_level: 2
2022-07-04T09:37:35.046661230Z -- gradient_support_level: 2
2022-07-04T09:37:35.046665764Z -- initial_point_support_level: 3
2022-07-04T09:37:35.046672463Z -- options: {'maxiter': 100, 'disp': False, 'ftol': 1e-06, 'eps': 1.4901161193
2022-07-04T09:37:35.046677546Z -- max_evals_grouped: 1
2022-07-04T09:37:35.046683473Z -- kwargs: {'tol': None}
2022-07-04T09:37:35.046689198Z ==============================================================
```

One of the job ids about this : cb1fncea4p74378e70v0

Steps to reproduce
The code to reproduce the error :

```
import qiskit
from qiskit.opflow import Z
from qiskit.circuit.library import TwoLocal
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit.algorithms.optimizers import COBYLA

service = QiskitRuntimeService(channel='ibm_quantum')

n = 2
operator = Z^Z
ansatz = TwoLocal(n, 'ry', 'cx', 'linear', reps=1)

runtime_inputs_cobyla = {
    'ansatz': ansatz,
    'operator': operator,
    'optimizer': COBYLA(maxiter=1),
}

options = {
    'backend_name': 'ibmq_qasm_simulator'
}

job_cobyla = service.run(
    program_id='vqe',
    options=options,
    inputs=runtime_inputs_cobyla,
)
```

Expected behavior
The right optimizer shows in the logs

Suggested solutions

Additional Information

- **qiskit-ibm-runtime version:** 0.5.0
- **Python version:** 3.9.12
- **qiskit version:** 0.37

Assignees
No one assigned

Labels
bug priority: high

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
Fix independence of `SciPyOptimizer` objects
Qiskit/qiskit
Fix `Optimizer` reconstruction from `settings`
Qiskit/qiskit

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Participants

LenaPer added bug on Jul 4, 2022

rathishcholarajan on Jul 6, 2022 Member

@Cryoris Could you please help check?

LenaPer on Jul 7, 2022 Author

Weirdly enough without changing anything on my environment I notice now the optimiser is not ignored anymore, but I'm still seeing that when using any optimizer other than SPSA/QNSPSA, any parameters I pass to the optimizer gets ignored and it uses the default ones (looking at this via `job.logs()`)

LenaPer changed the title ~~VQE Runtime ignores any optimizer other than SPSA/QNSPSA~~ VQE Runtime ignores any parameters of optimizer other than SPSA/QNSPSA on Jul 7, 2022

Cryoris on Jul 8, 2022 Contributor

Yeah we updated the VQE program and alongside fixed that the optimizers are ignored 🙂 Ok the lost settings are certainly a bug though, not quite sure where this originates from since the optimizers have a `settings` attribute which the runtime JSON encoder and decoders should use to reconstruct the object...

LenaPer on Jul 8, 2022 Author

As an example, when I pass COBYLA(maxiter=1), it shows in the logs it still has the default parameters

```
2022-07-08T14:06:50.003629122Z Optimizer: COBYLA
2022-07-08T14:06:50.003629122Z -- method: cobyla
2022-07-08T14:06:50.003629122Z -- bounds_support_level: 1
2022-07-08T14:06:50.003629122Z -- gradient_support_level: 1
2022-07-08T14:06:50.003629122Z -- initial_point_support_level: 3
2022-07-08T14:06:50.003629122Z -- options: {'disp': False, 'maxiter': 1000, 'rhobeg': 1.0}
2022-07-08T14:06:50.003631508Z -- max_evals_grouped: 1
2022-07-08T14:06:50.003636219Z -- kwargs: {'tol': None}
```

LenaPer on Jul 13, 2022 Author

Any updates on this?

Cryoris on Jul 13, 2022 Contributor

Not yet -- I'll try having a look this week 😀

Cryoris on Jul 14, 2022 · edited by Cryoris Edits Contributor

This issue seems to come from the encoding and decoding the optimizer which reset the settings. For example, the following serialization and deserialization of COBYLA

Goo

```
import json
from qiskit.algorithms.optimizers import COBYLA
from qiskit.providers.ibmq.runtime import RuntimeEncoder, RuntimeDecoder

obj = COBYLA(maxiter=1)
print("Original:\n", obj.settings)
dumped = json.dumps(obj, cls=RuntimeEncoder)
loaded = json.loads(dumped, cls=RuntimeDecoder)
print("Loaded:\n", loaded.settings)
```

prints

```
Original:
 {'max_evals_grouped': 1, 'options': {'maxiter': 1, 'disp': False, 'rhobeg': 1.0}, 'tol': None}
Loaded:
 {'max_evals_grouped': 1, 'options': {'maxiter': 1000, 'disp': False, 'rhobeg': 1.0}, 'tol': None}
```

It seems this comes from the problem that SciPy-based optimizers cannot be reconstructed from their own settings

```
>>> COBYLA(**obj.settings).settings
{'max_evals_grouped': 1, 'options': {'maxiter': 1000, 'disp': False, 'rhobeg': 1.0}, 'tol': None}
```

which is an assumption we have with the runtime encoder and decoders.

Cryoris on Jul 14, 2022 · edited by Cryoris Edits Contributor ···

Ok something really funky is going on (keep track of `maxiter`)

```
>>> from qiskit.algorithms.optimizers import COBYLA
>>> one_iter_optimizer = COBYLA(maxiter=1)
>>> one_iter_optimizer.settings
{'max_evals_grouped': 1, 'options': {'maxiter': 1, 'disp': False, 'rhobeg': 1.0}, 'tol': None}
>>> COBYLA(**one_iter_optimizer.settings)
<qiskit.algorithms.optimizers.cobyla.COBYLA object at 0x7f82100709d0>
>>> one_iter_optimizer.settings
{'max_evals_grouped': 1, 'options': {'maxiter': 1000, 'disp': False, 'rhobeg': 1.0}, 'tol': None}
```

Cryoris mentioned this on Jul 14, 2022

Fix independence of `SciPyOptimizer` objects Qiskit/qiskit#8343

Cryoris on Jul 14, 2022 Contributor ···

Qiskit/qiskit/pull/8343 (plus backport and update of the runtime images) should fix this 🙂

Cryoris mentioned this on Jul 19, 2022

Fix `Optimizer` reconstruction from `settings` Qiskit/qiskit#8381

rathishcholarajan added priority: high on Jul 20, 2022

kt474 on Feb 14, 2023 Member ···

This issue has been resolved

Fix `Optimizer` reconstruction from `settings` #8381

<> Code

Merged mergify merged 4 commits into `Qiskit:main` from `Cryoris:fix-optimizer-settings` on Jul 20, 2022

Conversation 2 | Commits 4 | Checks 0 | Files changed 3 | +57 −5

Cryoris commented on Jul 19, 2022 • edited — Contributor

Summary

Fix `Optimizer` reconstruction from `settings`. Fixes Qiskit/qiskit-ibm-runtime#407 and supersedes #8343 for the bugfix release.

Also fixes #6682 where this has been described a while ago.

Details and comments

Fix a bug where creating a new instance of a SciPy optimizer has overwritten the settings of a previously created instance of the same type. For example, previously:

```
from qiskit.algorithms.optimizers import COBYLA
optimizer = COBYLA(maxiter=1)
reconstructed = COBYLA(**optimizer.settings)
```

Created a new `COBYLA` instance with the wrong value `maxiter=1000` (= the default value) and *additionally* reset the iterations of the original `optimizer` instance to 1000 as well.

Cryoris added 3 commits 3 years ago

- `fix settings in optimizers` — Unverified — aec759f
- `add reno` — Unverified — 66778d4
- `don't duplicate args` — Unverified — e1e06c9

Cryoris requested review from **a team**, **manoelmarques** and **woodsp-ibm** as code owners 3 years ago

qiskit-bot commented on Jul 19, 2022 — Collaborator

Thank you for opening a new pull request.

Before your PR can be merged it will first need to pass continuous integration tests and be reviewed. Sometimes the review process can be slow, so please be patient.

While you're waiting, please feel free to review other open PRs. While only a subset of people are authorized to approve pull requests for merging, everyone is encouraged to review open pull requests. Doing reviews helps reduce the burden on the core team and helps make the project's code better for everyone.

One or more of the the following people are requested to review this:

- @Qiskit/terra-core
- @manoelmarques
- @woodsp-ibm

Reviewers

woodsp-ibm

manoelmarques

Assignees

No one assigned

Labels

Changelog: Bugfix | mod: algorithms | stable backport potential

Projects

None yet

Milestone

0.21.1

Development

Successfully merging this pull request may close these issues.

VQE Runtime ignores any parameters of opti...

`ScipyOptimizer` options are overriden by de...

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3 participants